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Executive summary

Europe's policy options for a dynamic and trustworthy development of the Internet of Things

SMART 2012/0053

Schindler HR, Cave J, Robinson N, Horvath V, Hackett P, Gunashekar S, Botterman M, Forge S, Graux H

Final Report (D7)

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Prepared for the European Commission, DG Communications Networks, Content and Technology (CONNECT)

Executive summary

The Internet of Things (IoT) is developing rapidly and may challenge conventional business, market, policy and societal models. In particular, the economic, socio-political, legal and technological governance of the internet is based on assumptions about rational choice, market forces and effective self-organisation that are most appropriate to human-controlled systems. The interacting autonomous systems of the IoT are already departing from this paradigm. This raises two complementary policy challenges: whether these departures raise any unique IoT-specific policy concerns, let alone specific problems in need of legal intervention or policy support; and whether the development of the IoT affects the rationale, impacts and/or available instruments for existing interventions ranging from regulation to development and deployment support.

A study to inform a consistent policy stance towards the IoT

This report commissioned by the European Commission aims to inform the development of a consistent European policy stance capable of fostering a dynamic and trustworthy IoT that helps meet key European challenges.

The study addresses the following research question:

What can usefully be done to stimulate the development of the Internet of Things in a way that best supports Europe's policy objectives (societal impact and jobs through innovation), while respecting European values and regulations (with particular reference to ethics and data protection)?

The study builds on European Commission work initiated in 2005, including policy discussions and recommendations including those of the European Commission's IoT

Expert Group (2010-2012) in particular, the six challenges (identification, privacy and data protection and security, architectures, ethics, standards and governance) identified by that group. We also build on the results of the 2012 public consultation on the IoT conducted in the second quarter of 2012 (European Commission, 2013).

The study was informed by a literature review, key informant interviews, and a team-internal scenarios-based workshop. It also extended and tested its findings and conclusions at an open stakeholder workshop held on 30 April 2013 at the European Commission's premises in Brussels.

Results and findings

A helpful starting definition

Building on the definition given by the International Telecommunication Union (ITU), the study proposes the following definition of the IoT:

> The Internet of Things builds out from today's internet by creating a pervasive and self-organising network of connected, identifiable and addressable physical objects enabling application development in and across key vertical sectors through the use of embedded chips.[1]

The IoT presents issues across several domains

According to experts interviewed for this study, **the current development of the IoT may not be aligned with Europe's policy objectives. Part of this is due to the limited influence of European (government and industry) actors and it may not be possible to address the consequences once the IoT has matured.**

The socioeconomic impacts of the IoT are expected by many industry analysts to develop **rapidly over the next five to ten years into an important element of the European digital economy**. But it cannot be assumed that this growth will be coherent or manageable. Current trajectories suggest the emergence of multiple competing architectures and identification schemes, leading to potentially damaging **fragmentation** across and within sectors or the triumph of a **'second best'** candidate. This is **not simply a market phenomenon**; the governance of the internet involves standardisation, government policy and a measure of self-regulation, but the **institutions and decisions may lack accountability** and may not effectively balance competing interests.

In order for government, business and societal organisations to realise the potential of the IoT and meet its challenges, its application must be accepted and trusted – not universally, uncritically or unequivocally, but in a proportionate, reasoned and effective manner. This requires both accurate and comprehensible information and the operationalisation of ethical principles.

[1] Integrated circuits or microprocessors, commonly called 'chips'.

Some levels of the IoT value chain are likely to suffer classic market failure due to entry and exit barriers, natural monopolies, information asymmetries and externalities. This will not be true of all layers – the openness and interoperability on which the IoT business proposition rests should lower barriers in the device layer. But this, too, can be problematic, if effective use depends on continuing collaborative innovation by end users, application providers and other stakeholders, and if switching is easier than (investment in) adaptation.[2] The unreliability of unaided market forces can be seen, for example, in inadequate investment in security (externalities), the unreliability of consumer sovereignty (information asymmetry) and the barriers to innovation arising from inappropriate spectrum policy and fragmentary or closed standards.

State of play: the growing potential of the IoT

Section 1 examines the context in which the IoT is emerging. Despite its youth, **the IoT is seen as one of the fastest growing IT segments**. By 2020, upper estimates of its annual global economic potential[3] across all affected sectors range from $1.4 trillion to $14.4 trillion. Some of the most promising and intriguing opportunities come from the linkage of the IoT to other systems and technologies, such as clouds, smart grids, nanotechnology and robotics.

Key issues: wide-ranging effects on competition and competitiveness

The potential economic implications of a rapidly developing IoT reflect the changing pattern of **horizontal and vertical** relations among the businesses that supply, use and serve the IoT. Of particular interest is the **competitive tension** between large market players from these other sectors, and the potential for a more open environment for small and medium-sized enterprises (SMEs) and innovative entrants within and beyond the IoT. In particular, because the 'things' of the IoT act autonomously and as part of a densely linked ecosystem, sole control of the IoT cannot be assumed to lie with the owners of devices or with providers of essential infrastructure services.

As in other emerging technology domains, European competitiveness can be enhanced by exploiting **the strength of its research capabilities**, reinforced by **antitrust**, **public procurement and international trade initiatives**. More specifically, businesses and other IoT users can benefit from the embedding of existing European **standards of consumer and data protection**, which may become a unique source of competitive advantage in marketing European IoT technologies and services worldwide.

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[2] This excess volatility (Katz and Shapiro, 1992) is characteristic of markets with network externalities and has been noted as well in the 'app economy' (Cave et al., 2012).

[3] Taking into account machine-to-machine (M2M) and Metropolitan Mesh Machine Network (M3N) applications.

But this depends on the availability and structure of **investment**. China has already earmarked €625m for IoT investment. Public and private sector investors will have the possibility to provide infrastructure and application funding for the IoT including private financing, public investment, public–private partnerships, and social (eg crowd-funded) finance.

Building an ethical IoT

The commercial and technical development and broader socioeconomic impacts of any information and communications technology (ICT) derive from the way it deals with the ethical tensions arising from the way it connects people and organisations with different objectives. The IoT creates new forms of contact that make it hard for those currently charged with responsibilities to know, understand and control these connections; the classical protections of negotiation, markets and contracts may not work as well for human-to-machine, let alone M2M contacts. Having investigated these ethical issues, especially in relation to **privacy**, **autonomy**, **trust**, **identity and social inclusion**, we note that without greater attention to improving individual understanding and awareness, solutions, even if implemented, might not survive.

Architecture, security and identification

The challenges of the IoT architecture, identification and security are examined in more depth in Section 4. Architecture, in particular, must provide a set of common rules to keep systems close enough to allow interoperability and thus facilitate efficient emergence of better systems, while allowing enough flexibility to encourage innovation.

Current development is producing substantial heterogeneity of applications, environments and systems but not (seamless) interoperability. The resulting technical and cost issues across sectors make it likely that a range of specialist **architectures** will emerge that can lock in this weakness.

The same fragmentation risk affects naming and addressing norms; differences across geographical areas and industries can limit interoperability and competition.

Much of this fragmentation can be overcome by suitable open standards. Many of the applicable standards are inherited from other areas (eg radio communication, general ICT); there is thus an ongoing debate over the necessity for IoT-specific standards and the role of specific standards bodies. Several specific initiatives have been created to incorporate and adapt existing standards and to complete the IoT portfolio with additional standards. However, at this early stage of development, it is not clear how uniformly standards are applied or enforced, nor have any preferable approaches been identified for application and enforcement.

The ubiquity of sensor (and eventually actuator) networks poses some unique and interesting questions with respect to security as a public good and whether **cyber security** is subject to market failure as well as adding novel aspects to pre-existent privacy and

security risks. The IoT could also result in cyber-attacks targeting new endpoints, such as smart homes, therefore requiring strategies that can be efficient across multiple domains and competing priorities.

Strategic objectives of European IoT governance

The study considered the case for action **at Community level and found arguments for its necessity and European added value**. Section 6 identifies the general objectives of such action. These include **accountability, safety and interoperability of an inclusive, ethical and open IoT**, characterised by **effective and efficient competition and competitiveness**.

Potential gaps in the existing legal framework

As an extension to the internet, the IoT is affected by internet governance structures. But the internet itself does not 'fit' neatly within all the relevant governance frameworks (telecom, competition, privacy, consumer protection and so on), Its problems are not always effectively addressed by extensions of these frameworks. And that extension may in turn weaken the effectiveness of existing competition and other rules. The IoT poses its own unique challenges and also involves additional reverence governance domains (eg safety, transportation) more centrally than the internet does. Section 7 examines relevant areas of EU law and identifies gaps that could hamper the realisation of the strategic objectives of IoT policymaking.

In particular, the frameworks for **competition** (in particular market definitions and the role of competition authorities), **privacy and data protection** (in particular regarding liability and responsibility), **universal service** and **cyber security** are likely to require adjustment or compensating 'soft law' measures.

Policy options

The formulation of concrete and specific policy options is possibly premature and went beyond the scope of the present study. In Section 8 we analyse three options for a broad policy approach and some tools for their operationalisation, which are summarised in Table 0.1.

Table 0.1 Summary of broad policy options

Option	EC activity	Efficiency	Efficacy
No action	Current trajectories continue	No guarantee for development in accordance with EU objectives	Market players retain complete freedom
Soft law	Using monitoring, innovation policy, industrial policy	If sufficient incentives for adoption and uptake exist, high effectiveness is possible, while incentivising coherence with EU policy objectives	Market players retain some freedom in deciding the most effective manner of complying with requirements
Hard law	Harmonisation and enforcement in IoT-related areas (e-commerce, data protection etc)	Depending on enforcement, mandatory compliance can be highly efficient	Negative externalities are hard to foresee given the early stage of technology development, therefore are difficult to avoid in legislation

Policy recommendations

Comparison of the effectiveness, efficiency and coherence of these high-level policy stances supports the soft law option at least in the near term as the best way to **create space** for IoT development, **accelerate or improve the development of the IoT market** and make progress in meeting the challenges. The details are developed as a set of recommendations in Section 9.

Our recommendations include a central role for the European Commission in **coordinating policy dialogue** to ensure common understanding and coherent effective action across sectors, regions and policy areas; support for **meaningful digital literacy programmes** and **awareness-raising** to empower self-regulation and improve individual interaction with the IoT; and support and promotion of **knowledge sharing**, **research and validation projects** with funding, **continuous debate and policy articulation** especially on identification, privacy and ethics in IoT environments. Although an **ethical charter** may be a useful component of self-regulation, support for the general approach is patchy. As an interim measure, **creating a European 'Ethical Tech' brand** could encourage innovators and providers to develop ethical technology in line with market and user needs.

Monitoring and implementation

The information necessary to track the development of the IoT is fragmented and difficult to use. This problem of too much, too little *and* the wrong kind of information may account for slow progress to date. Rapid changes – and the exaflood of data – are expected to continue. In order to develop and implement appropriate flexible and future-proof policy this problem must be overcome. In this regard, it is fortunate that the various governance domains that affect and are affected by the IoT each collect and analyse information. This creates the basis for a coordinated policy approach allied to an articulated information structure organised around the IoT. This **IoT observatory** could follow the impact of challenges in areas that cannot be directly measured, such as ethics, privacy and security, and should lead to more joined-up policy and deeper and more balanced understanding; the sharing of information and co-creation of data resources for monitoring and evaluation will also create shared understanding and help to break down organisational stovepipes. This is not limited to past and present information: **the information should be used to create a rich set of shared scenarios for joint explorations of the emerging IoT**. As always, **it is important to establish measure-specific indicators that underpin a monitoring strategy** (in conjunction with DG CONNECT's 'Metrics' initiative.